

WT LAB- Exp 4

Aim: Design and Implementation of a Rain Sensor Module with Clothes Protection Using Arduino Uno

Introduction: A Rain Sensor Module with Clothes Protection is an automated system designed to detect rain and take action to protect clothes hanging outside. The system uses an Arduino Uno as the central controller, a rain sensor module to detect water droplets, and a servo motor to control a protective cover. Upon detecting rain, the Arduino triggers the servo motor to move a cover over the clothes, preventing them from getting wet. This project integrates sensor input, motor control, and real-time decision-making, providing a practical application in home automation.

Arduino Uno Overview

The Arduino Uno is a widely used microcontroller board, valued for its ease of use, affordability, and flexibility. It features:

- **Microcontroller:** ATmega328P
- **Operating Voltage:** 5V
- **Digital I/O Pins:** 14 (6 PWM outputs)
- **Analog Input Pins:** 6
- **Flash Memory:** 32 KB
- **Clock Speed:** 16 MHz
- **USB Connection:** For programming and communication
- **Power Supply:** Can be powered via USB or an external source

The Arduino Uno is programmed using the Arduino IDE, allowing users to write and upload code to control connected electronic components.

Components Used in This Experiment

1. Arduino Uno

- Acts as the main processing unit.
- Processes sensor readings and controls the servo motor.
- Makes decisions based on real-time sensor input.

2. Rain Sensor Module

- Detects rain by sensing water droplets on its surface.
- Outputs a signal to the Arduino when moisture is detected.
- Comprises an exposed plate that changes resistance when wet.

3. Servo Motor (SG90)

- Controls the movement of a protective cover.
- Moves the cover over the clothes when rain is detected.
- Returns to the original position once the rain stops.

4. Protective Cover

- A waterproof sheet or shield that moves over the clothes.
- Prevents rain from wetting the clothes.
- Attached to the servo motor for automated movement.

How the Rain Sensor Module Works

1. Sensor Input

- The rain sensor detects water droplets and sends a signal to the Arduino.
- The Arduino reads the sensor's analog or digital output to determine moisture levels.

2. Decision Making

- If rain is detected, the Arduino sends a signal to activate the servo motor.
- The servo motor moves the protective cover over the clothes.
- If no rain is detected, the cover remains in its original position.

3. Motor Control

- The servo motor operates based on Arduino signals:
 - **Move Cover Over Clothes:** Rotates to a specific angle.
 - **Retract Cover:** Returns to the default position when the rain stops.

4. Continuous Loop

- The system continuously monitors the rain sensor and adjusts the cover accordingly.
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Simple Operational Flow

1. Start:

- System initializes; Arduino and sensors power on.

2. Detect Rain:

- Rain sensor checks for water droplets.
- Arduino reads sensor data.

3. Decision Making:

- **If rain is detected** → Activate servo motor to move the cover over the clothes.
- **If no rain is detected** → Keep the cover in its original position.

4. Repeat:

- Continuously monitor the rain sensor and react accordingly.

CODE:

```
#include <Servo.h>
```

```
// Define the pin for the rain sensor
```

```
const int rainSensorPin = A0; // You can change this pin as per your wiring
```

```
// Define the pin for the servo motor
```

```
const int servoPin = 8; // You can change this pin as per your wiring
```

```
Servo myServo;
```

```
void setup() {
```

```
  // Set up the rain sensor pin as an input
```

```
  pinMode(rainSensorPin, INPUT);
```

```
// Set up the servo motor

myServo.attach(servoPin);

}

void loop() {

  // Read the state of the rain sensor

  int rainState = digitalRead(rainSensorPin);

  // If the rain sensor detects rain (LOW), rotate the servo motor continuously

  if (rainState == LOW) {

    myServo.write(180); // Rotate the servo motor to 90 degrees

  } else {

    // If no rain is detected, stop the servo motor

    myServo.write(0); // Rotate the servo motor to 0 degrees

  }

  // Add a small delay for stability

  delay(100);

}
```

Code Breakdown:

1. Pin Setup

- The rain sensor pin is set as an input.
- The servo motor pin is attached to control its movement.

2. Detect Rain

- The Arduino reads the rain sensor's output to check for moisture.
- The sensor provides a **LOW** signal when rain is detected and **HIGH** when no rain is present.

3. Decision Making

- **If rain is detected (LOW signal):** The servo motor rotates to **180 degrees** to move the protective cover over the clothes.
- **If no rain is detected (HIGH signal):** The servo motor rotates back to **0 degrees** to retract the cover.

4. Motor Control

- **moveCover()** → Moves the cover over the clothes when rain is detected.
- **retractCover()** → Returns the cover to its original position when no rain is detected.

5. Continuous Loop

- The Arduino continuously checks the sensor and adjusts the cover based on real-time readings.
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Conclusion

The Rain Sensor Module with Clothes Protection using Arduino Uno is an efficient and automated solution for protecting clothes from unexpected rain. It demonstrates the integration of sensors, motors, and real-time decision-making, making it a practical project for home automation applications.

